

Scientific Reliability and Modern AI in Astrophysics

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IAG-USP | Postgraduate mini-course

Today's structure

Slot 1: scientific reliability

- What makes astrophysical data difficult?
- Bias, leakage, imbalance, domain shift
- Uncertainty, calibration, interpretability
- Reliability checklist for ML claims

Slot 2: modern AI in Astrophysics

- Transfer and self-supervised learning
- Foundation models and multimodal learning
- Generative AI and diffusion models
- Scientific ML and simulation-based inference

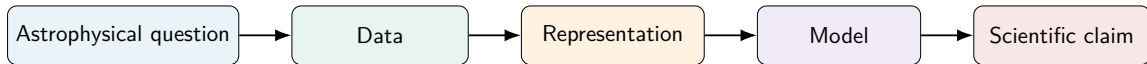
Main question: when should we trust an AI result as astrophysics?

Learning outcomes

By the end of this lesson, you should be able to:

- ① identify common failure modes in astrophysical ML studies;
- ② distinguish measurement noise, label noise, selection effects, and domain shift;
- ③ explain why accuracy is often insufficient for scientific use;
- ④ interpret calibrated probabilities and uncertainty estimates;
- ⑤ evaluate whether a model has learned physics, provenance, or shortcuts;
- ⑥ describe how transfer learning, self-supervision, foundation models, and diffusion models enter modern astrophysics.

From Monday and Wednesday to Friday



Today's focus

The hard part is often not building the model. The hard part is defending the arrow from **model output** to **astrophysical conclusion**.

Why reliability matters in Astrophysics

Astrophysical ML results often influence:

- candidate lists for expensive spectroscopic follow-up;
- catalog construction and population studies;
- physical parameter inference;
- anomaly discovery;
- claims about rare or new classes of objects;
- downstream cosmology or stellar-evolution conclusions.

Core warning

A model can be numerically accurate while scientifically misleading.

The textbook assumption: i.i.d. data

Many learning algorithms implicitly assume:

$$(\mathbf{x}_i, y_i) \sim P_{\text{train}}(\mathbf{x}, y) , \quad (\mathbf{x}_j, y_j) \sim P_{\text{test}}(\mathbf{x}, y) ,$$

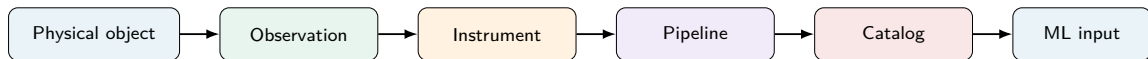
with

$$P_{\text{train}}(\mathbf{x}, y) \approx P_{\text{test}}(\mathbf{x}, y).$$

But astronomical data are often not i.i.d.

Objects may differ by survey, cadence, filter system, sky region, exposure depth, selection function, reduction pipeline, instrument, or label source.

The observational pipeline is part of the data



- The model never sees the object directly.
- It sees a processed representation of the object.
- Every step can introduce systematic structure.

Reliability principle

Before asking “what did the neural network learn?”, ask: **what information was available to learn?**

A taxonomy of difficulty

Difficulty	Examples in Astrophysics
Measurement noise	flux uncertainties, read noise, sky background, polarization errors
Irregular sampling	seasonal gaps, sparse cadence, heterogeneous survey visits
Missingness	non-detections, bad pixels, masked spectral regions
Selection effects	magnitude limits, color cuts, follow-up targeting biases
Label problems	uncertain classes, human labels, literature compilation biases
Domain shift	train on one survey/cluster/instrument, apply to another
Physical degeneracy	different physical states producing similar observables

Measurement uncertainty is not decoration

For a light curve or spectrum, a realistic data point is not just

$$(t_j, f_j) ,$$

but often

$$(t_j, f_j, \sigma_j, m_j) ,$$

where σ_j is a measurement uncertainty and m_j is a mask or quality flag.

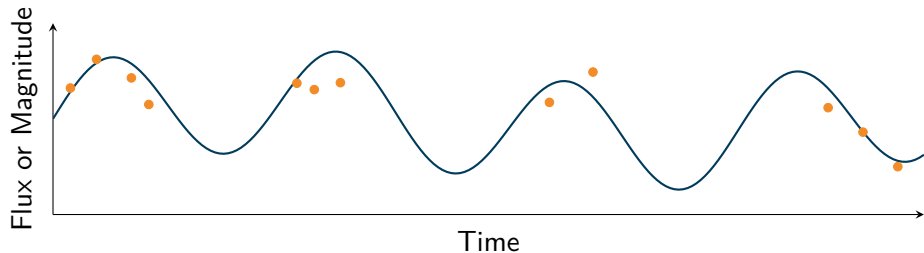
Homoscedastic simplification

$$f_j = f_{\text{true}}(t_j) + \epsilon_j , \quad \epsilon_j \sim \mathcal{N}(0, \sigma^2) .$$

Astrophysical reality

$$f_j = f_{\text{true}}(t_j) + \epsilon_j , \quad \epsilon_j \sim \mathcal{N}(0, \sigma_j^2) .$$

Irregular sampling changes the learning problem



- A sequence model may need Δt_j , masks, uncertainty channels, or positional encodings.
- Resampling can create artifacts if it pretends that unobserved values were measured.
- Cadence itself may become a shortcut if different classes come from different observing programs.

Label noise and label semantics

In supervised learning, the label is often treated as ground truth:

$$y_i \in \{0, 1\} .$$

But in astrophysics, labels may be:

- human classifications with disagreement;
- literature labels from heterogeneous criteria;
- proxy labels from color cuts or variability criteria;
- coarse labels that hide physical subtypes;
- temporally unstable labels: an object may change state.

Scientific risk

If labels are physically heterogeneous, the model may learn the label-construction process rather than the astrophysical class.

Example: multi-resolution labels

A common problem: two datasets encode the same phenomenon at different semantic resolution.

Dataset	Raw label	Derived task label
A	non-Be	non-Be
A	Be	Be, subtype unknown
B	non-Be	non-Be
B	pole-on Be	Be, pole-on
B	edge-on Be	Be, edge-on
C	B-color candidate	selection flag, not class label

Best practice

Preserve raw labels and label provenance before defining task-specific targets.

Class imbalance: the accuracy trap

Suppose a candidate catalog contains:

$$N_{\text{ordinary}} = 98,000 , \quad N_{\text{rare}} = 2,000 .$$

A classifier that always predicts “ordinary” has

$$\text{accuracy} = 98\% .$$

Looks good

High accuracy.

Scientifically useless

Zero rare objects recovered.

For rare-object astrophysics, use precision, recall, PR curves, MCC, and threshold-aware evaluation.

Selection functions: data are collected with intent

The observed sample is not usually a random sample from the Universe.

Schematic

$$P_{\text{observed}}(\mathbf{x}, y) \propto P_{\text{universe}}(\mathbf{x}, y) S(\mathbf{x}, y, \Omega, t, \text{instrument}) ,$$

where S is the selection function.

Examples:

- spectroscopic labels exist preferentially for bright targets;
- rare candidates may be overrepresented in follow-up samples;
- color cuts define a physically meaningful but incomplete population;
- sky coverage and extinction vary spatially.

Data leakage: when the answer enters the input

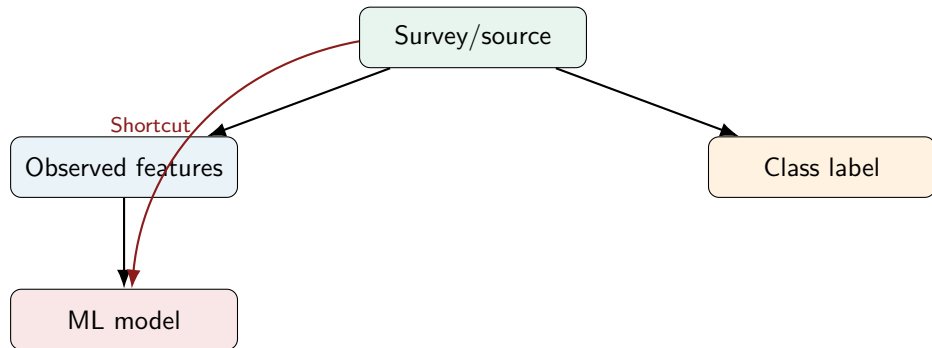
Data leakage

A model uses information during training or evaluation that would not be available in the intended scientific deployment.

Common examples:

- the same physical object appears in both train and test sets;
- multiple observations of the same object are split independently;
- preprocessing is fit on all data before the train/test split;
- labels are encoded indirectly through filenames, source flags, or selection masks;
- duplicate or near-duplicate light curves cross partitions;
- hyperparameters are repeatedly tuned on the test set.

Domain shift and shortcut learning



Diagnostic question

Can a simple model predict the survey/source from the same features used for the science task?

Types of distribution shift

Shift type	Meaning
Covariate shift	$P_{\text{train}}(\mathbf{x}) \neq P_{\text{test}}(\mathbf{x})$ but $P(y \mathbf{x})$ similar
Label shift	$P_{\text{train}}(y) \neq P_{\text{test}}(y)$
Concept shift	$P_{\text{train}}(y \mathbf{x}) \neq P_{\text{test}}(y \mathbf{x})$
Domain shift	train and deployment come from different surveys, instruments, fields, or epochs
Out-of-distribution	deployment includes objects not represented in training data

Astrophysical deployment

The intended use case must define which shift matters most.

Metrics must match the scientific use case

Scientific use	False positive cost	False negative cost
Follow-up candidate list	high	moderate
Rare-object discovery	moderate	high
Clean reference catalog	high	high
Exploratory triage	low/moderate	moderate
Alert broker filtering	time-dependent	time-dependent

Thresholds are scientific decisions

A probability threshold of 0.5 is rarely special. Choose thresholds on validation data according to the intended use.

Probability calibration

A classifier is calibrated if, among objects assigned probability p , approximately a fraction p truly belong to the class:

$$P(Y = 1 \mid \hat{p} = p) \approx p .$$

Good ranking

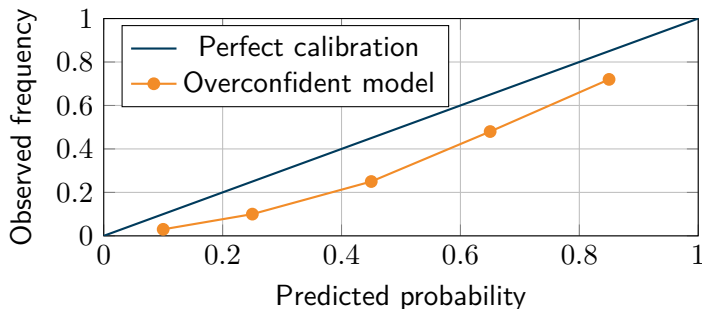
The most likely candidates appear at the top.

Good calibration

A score of 0.8 behaves like an 80% probability.

A model can rank well but be badly calibrated.

Reliability diagram: what to inspect



Interpretation

If observed frequencies lie below the diagonal, the model is overconfident.

Uncertainty: two broad sources

Aleatoric uncertainty

Intrinsic noise or ambiguity in the data-generating process.

- photon noise
- weak signal-to-noise
- physically degenerate observables
- stochastic variability

Epistemic uncertainty

Uncertainty due to limited knowledge or training data.

- sparse labels
- incomplete training set
- model uncertainty
- domain mismatch

Aleatoric uncertainty is not solved by more data of the same quality; epistemic uncertainty often is.

Practical uncertainty tools

Tool	What it gives
Bootstrap	empirical confidence intervals for metrics or predictions
Deep ensembles	model disagreement and often strong predictive uncertainty
Monte Carlo dropout	approximate stochastic prediction distribution
Quantile regression	intervals for continuous targets
Bayesian neural nets	posterior distribution over weights or predictions
Conformal prediction	finite-sample prediction sets under exchangeability assumptions

Caution

Uncertainty estimates should be validated. A confidence interval is not trustworthy just because a model outputs one.

Interpretable does not automatically mean physical

Interpretability tools can help diagnose model behavior:

- permutation importance;
- feature importance in tree models;
- SHAP values;
- saliency maps;
- occlusion or ablation tests;
- attention visualization;
- embedding projections.

Important distinction

Interpretability methods explain model sensitivity, not necessarily physical causality.

Model cards for astrophysical ML

For every serious model, record:

- scientific task and intended deployment;
- training data provenance and selection function;
- label definitions and uncertainty;
- preprocessing and feature representation;
- split strategy and leakage checks;
- metrics and threshold policy;
- calibration status;
- known failure modes;
- out-of-domain restrictions;
- reproducibility information.

Case study logic: Be-star photometric classification

A scientifically reliable Be-star classifier should ask:

- 1 Are source labels, cluster labels, and raw labels preserved?
- 2 Are broad Be, subtype, and B-color-candidate labels kept distinct?
- 3 Are train/test splits object-level and leakage-safe?
- 4 Can the model predict survey identity from the features?
- 5 Do metrics include recall, precision, PR-AUC, MCC, and per-source confusion matrices?
- 6 Are probabilities calibrated before candidate ranking?
- 7 Are false positives and false negatives interpreted astrophysically?

Micro-exercise: would you trust this result?

A paper reports:

- 96% accuracy for classifying rare transients;
- train/test split was random over detections, not physical objects;
- no precision-recall curve;
- labels compiled from follow-up catalogs;
- no per-survey confusion matrix;
- no calibration analysis.

Discussion questions

- ① What are the main reliability risks?
- ② Which missing diagnostic would you request first?
- ③ What claim could still be made safely?

Slot 1 summary

- ➊ Astrophysical ML is vulnerable to noise, selection effects, label ambiguity, leakage, and domain shift.
- ➋ Evaluation must match the intended scientific use case.
- ➌ Calibration and uncertainty are essential when model scores guide follow-up or catalog construction.
- ➍ Interpretation tools are useful diagnostics, but not automatic physical explanations.
- ➎ Reproducibility and provenance are part of the scientific result.

Modern AI: what changed?

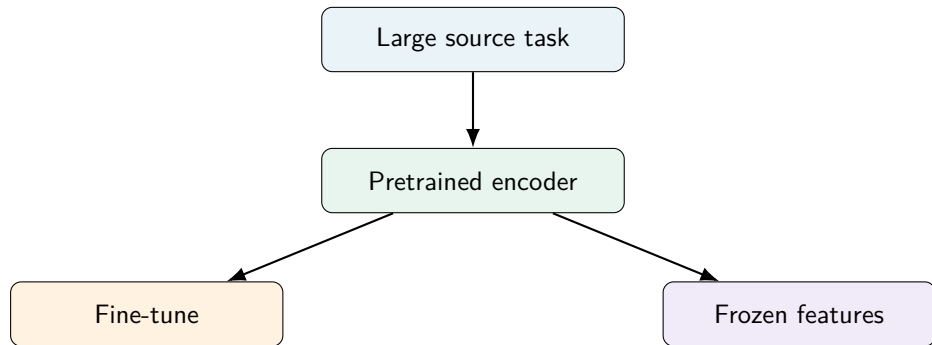
Classical supervised learning asks:

$$\mathcal{D}_{\text{labeled}} \longrightarrow f_{\theta}(\mathbf{x}) \approx y .$$

Modern AI often changes one or more ingredients:

- learn representations before labels are available;
- transfer knowledge across tasks, surveys, or modalities;
- train on very large unlabeled datasets;
- generate plausible data samples;
- emulate expensive simulations;
- combine observations with physical constraints.

Transfer learning



Astrophysical use

Use a representation learned from abundant data, then adapt it to a smaller scientific sample.

Self-supervised learning

Self-supervised learning creates supervision from the data itself.

Examples of pretext tasks

- mask part of a spectrum and reconstruct it;
- predict missing light-curve points;
- contrast two augmented views of the same object;
- order or align time-series segments;
- learn invariance across filters or surveys.

Why useful in astronomy?

- labels are expensive;
- unlabeled survey data are abundant;
- representations can transfer;
- rare-class performance may improve.

Foundation models in astronomy

Working definition

A foundation model is a large pretrained model whose learned representation can be adapted to many downstream tasks.

Astronomical examples now include models for:

- light-curve embeddings and few-shot classification;
- multimodal time-series plus tabular classification;
- image and spectrum representation learning;
- cross-survey transfer.

Examples include ATAT for astronomical time series and tabular data, and Astromer-style models for light-curve embeddings and few-shot classification. See Cabrera-Vives et al. 2024; Donoso-Oliva et al. 2025/2026.

Recent survey scale increases the pressure

Modern surveys make manual inspection impossible at full scale.

Rubin Observatory / LSST

First-look images were publicly revealed in June 2025, marking a major step toward next-generation time-domain survey data.

Euclid

The first survey data release in March 2025 included deep-field data and tens of millions of detections for astrophysical studies.

Consequence

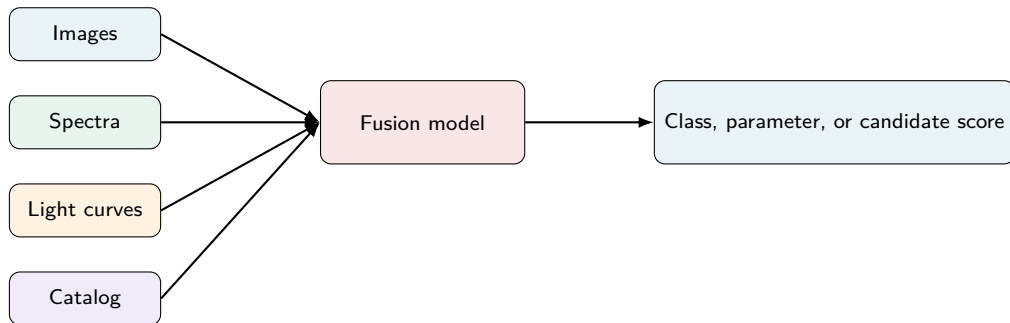
ML becomes infrastructure: classification, triage, anomaly detection, deblending, calibration, candidate ranking, and simulation acceleration.

Rubin Observatory First Look, 2025; ESA Euclid Q1 data release, 2025.

Multimodal learning

Astrophysical objects are often observed through multiple channels:

$$\mathbf{x} = (\mathbf{x}_{\text{image}}, \mathbf{x}_{\text{spectrum}}, \mathbf{x}_{\text{lightcurve}}, \mathbf{x}_{\text{catalog}}, \mathbf{x}_{\text{polarimetry}}) .$$



Key question

Does each modality add physical information, or does it add only survey/provenance information?

Generative models: why they matter

Generative models learn a distribution:

$$\mathbf{x} \sim p_{\theta}(\mathbf{x}) ,$$

or a conditional distribution:

$$\mathbf{x} \sim p_{\theta}(\mathbf{x} \mid \text{condition}) .$$

Astrophysical uses:

- image generation and augmentation;
- denoising and reconstruction;
- simulation emulation;
- posterior approximation;
- missing-data imputation;
- controlled synthetic benchmarks.

Scientific caution

A realistic-looking generated object is not automatically a physically valid object.

Diffusion models: the basic idea

Forward process: gradually add noise

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I) .$$

Reverse process: learn to remove noise step by step

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) .$$

Interpretation

A diffusion model learns how to move from noise toward samples that resemble the training distribution.

Diffusion models in astrophysical contexts

Examples of active or emerging use:

- radio image reconstruction from dirty images;
- galaxy image generation conditioned on morphology or physical properties;
- denoising interstellar dust or simulation fields;
- fast generation of cosmological fields;
- synthetic data for controlled ML stress tests.

Validation requirement

Evaluate generated data using physical statistics, not visual realism alone.

Examples include conditional diffusion for radio-astronomical image reconstruction and category/physics-conditioned galaxy image generation. See Drozdova et al. 2024; Zhao et al. 2023; GalCatDiff 2026.

Simulation-based inference

When likelihoods are difficult but simulations are available:

$$\theta \longrightarrow \text{simulator} \longrightarrow \boldsymbol{x} .$$

We want the inverse:

$$\boldsymbol{x}_{\text{obs}} \longrightarrow p(\theta \mid \boldsymbol{x}_{\text{obs}}) .$$

Neural approaches can learn:

- likelihood ratios;
- posterior distributions;
- summary statistics;
- fast emulators of expensive simulators.

Astrophysical applications

Cosmology, stellar populations, gravitational waves, galaxy formation, radiative-transfer models, and exoplanet atmospheres.

Scientific machine learning

Scientific ML combines data-driven learning with physical structure.

Examples

- physics-informed neural networks;
- neural differential equations;
- differentiable simulations;
- neural operators;
- surrogate models;
- symmetry/equivariance constraints.

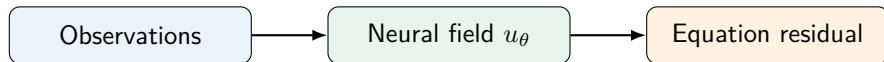
Astrophysical value

- faster parameter exploration;
- fewer unphysical predictions;
- interpretable structure;
- efficient simulation pipelines;
- better extrapolation when physics is correct.

Physics-informed neural networks: why include them?

Basic idea

A physics-informed neural network (PINN) is trained not only to fit data, but also to approximately satisfy known physical equations.



- Useful when data are sparse or noisy but the governing equations are partly known.
- Natural fit for PDE/ODE problems: fluids, stellar structure, radiative transfer, orbital dynamics, plasma/MHD models.
- They sit between pure data-driven ML and classical numerical simulation.

Canonical reference: Raissi, Perdikaris, and Karniadakis, J. Comput. Phys. 2019.

PINN formulation in one slide

Suppose a physical field $u(x, t)$ satisfies a differential equation

$$\mathcal{N}[u](x, t) = 0 ,$$

with initial/boundary conditions and possibly sparse observations.

A neural network represents the field:

$$u_{\theta}(x, t) \approx u(x, t) .$$

Automatic differentiation gives the residual:

$$r_{\theta}(x, t) = \mathcal{N}[u_{\theta}](x, t) .$$

The training objective combines data fit and physics violation:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \mathcal{L}_{\text{data}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{phys}} \mathcal{L}_{\text{residual}} .$$

Key message

The equation becomes part of the training signal!

Forward and inverse problems

Forward problem

Known physics and parameters:

$$\mathcal{N}[u; \theta_{\text{phys}}] = 0 .$$

Use the PINN to approximate the solution field:

$$(x, t) \mapsto u(x, t) .$$

Example: solve an ODE/PDE when a mesh-free differentiable approximation is convenient.

Inverse problem

Observed data, unknown physical parameter:

$$\mathcal{N}[u; \theta_{\text{phys}}] = 0 .$$

Learn both:

$$u_{\theta}(x, t) \quad \text{and} \quad \theta_{\text{phys}} .$$

Example: infer diffusion coefficient, opacity parameter, equation-of-state parameter, or source term.

PINNs are especially attractive when observations and equations constrain each other.

Astrophysical use cases

Equilibrium / structure

- Lane-Emden and stellar-structure equations
- hydrostatic equilibrium constraints
- Grad-Shafranov-like MHD equilibria
- cosmological background dynamics

Dynamics

- hydrodynamic and MHD fields
- shocks and transport equations
- orbital dynamics and gravitational potentials
- plasma and radiative-transfer models

Good teaching example

Start with a simple equation:

$$\frac{du}{dt} = -ku ,$$

and infer both $u(t)$ and k from sparse noisy data.

Why this belongs in Friday's lecture

It connects modern AI to scientific reliability: predictions are judged by data agreement *and* physical residuals.

What can go wrong with PINNs?

- **Loss balancing:** the data, boundary-condition, and residual terms can compete.
- **Scaling:** variables with very different units or magnitudes can dominate training.
- **Stiffness:** some equations are hard for standard optimization.
- **Discontinuities and shocks:** smooth networks may struggle with sharp fronts.
- **Boundary conditions:** weakly imposed boundaries may leak errors into the solution.
- **Identifiability:** inverse problems may have multiple physical solutions compatible with sparse data.
- **False confidence:** small residuals at collocation points do not automatically imply a globally accurate solution.

Caution

PINNs are not magic replacements for numerical solvers. Treat them as one scientific-ML tool whose assumptions must be validated.

When should an astrophysicist consider a PINN?

Use PINNs when...	Be cautious when...
there is a trusted differential equation	the governing physics is poorly specified
sparse data must be combined with known physics	labels are ordinary catalog classes
physical parameters are part of the inference	a standard solver is already faster and more reliable
constraints are more important than raw flexibility	shocks, turbulence, or multiscale stiffness dominate
mesh-free or differentiable solutions are valuable	validation data are absent or boundary conditions are uncertain

Course-level message

PINNs are not just “neural networks for physics”; they are a way of turning physical laws into an inductive bias and a validation criterion.

Micro-exercise: would you use a PINN?

For each case, decide whether a PINN is a natural first choice, a possible extension, or probably the wrong tool.

- 1 Classifying galaxies as spiral or elliptical from images.
- 2 Inferring stellar radius and mass from sparse observations plus stellar-structure equations.
- 3 Solving a simple radiative-transfer equation with unknown opacity from noisy measurements.
- 4 Detecting anomalous light curves in a survey with no known governing equation.
- 5 Reconstructing a smooth gravitational potential from stellar kinematic data.

Discussion point

The question is not “Can I add a physics loss?” but “Does the physics loss encode correct information for this scientific problem?”

Physics constraints and inductive bias

A neural network can be constrained by:

- symmetries: translation, rotation, permutation, time invariance;
- conservation laws;
- known equations or residual penalties;
- monotonicity constraints;
- physically motivated latent variables;
- observational uncertainty models.

Principle

Use domain knowledge to reduce the space of plausible functions.

Inductive bias is not a weakness; it is often the route to generalization.

Large language models in astronomy

LLMs may help with:

- literature triage;
- code generation and debugging;
- documentation and reproducibility templates;
- extracting structured information from papers;
- natural-language interfaces to data systems;
- hypothesis generation and teaching support.

Limitations

LLMs can hallucinate references, misstate equations, and produce plausible but wrong code. Treat them as assistants, not authorities.

Choosing a modern AI method

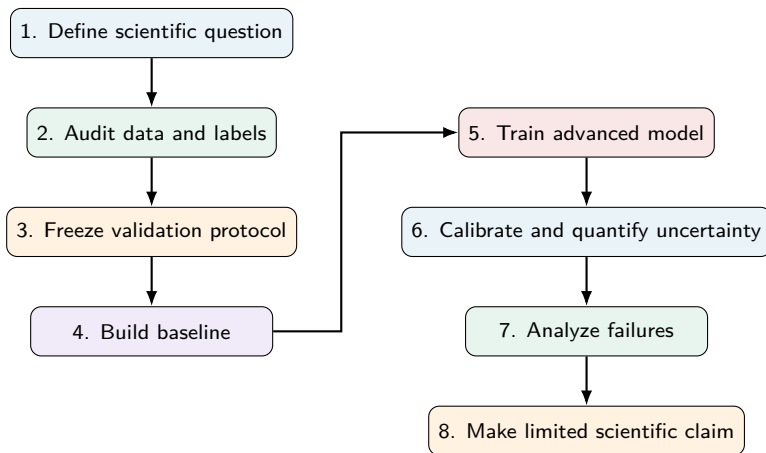
If the problem has...	Consider...
few labels, many unlabeled objects	self-supervised learning or transfer learning
many related tasks	foundation-style representations
multiple data modalities	multimodal fusion
expensive simulations	surrogate models or simulation-based inference
missing/noisy observations	probabilistic models, imputation, uncertainty-aware learning
need for synthetic controlled data	generative models or simulation augmentation
strong physical symmetries	equivariant or physics-informed architectures

Modern AI does not remove the reliability checklist

Even with foundation models or diffusion models, still ask:

- ① What distribution was the model trained on?
- ② What physical variables are missing?
- ③ What survey biases are encoded in the representation?
- ④ Does the benchmark match the deployment use case?
- ⑤ Are uncertainty and calibration evaluated?
- ⑥ Are generated or predicted outputs physically validated?
- ⑦ Can the result be reproduced from code and data instructions?

A reliability-first ML workflow



Micro-exercise: method selection

Choose a method and justify it.

- ① You have 2 million unlabeled light curves and 2,000 labeled rare objects.
- ② You have high-resolution simulated spectra but only a small observed labeled set.
- ③ You have dirty radio images and want reconstructed sky models.
- ④ You have multi-band photometry, spectra, and polarimetry for the same stars.
- ⑤ You need to infer posterior distributions for physical parameters from expensive simulations.

For each case: what is the main risk, and how would you validate the result?

Take-home messages

- ➊ Astrophysical ML is scientific inference under observational constraints.
- ➋ The data-generation process matters as much as the architecture.
- ➌ Reliability requires leakage control, domain-shift diagnostics, meaningful metrics, uncertainty, calibration, and error analysis.
- ➍ Modern AI expands the toolbox, but it does not remove the need for physical validation.
- ➎ The best project is not the one with the most complex model; it is the one with the strongest scientific argument.

Bridge to Week 2

Next week, the course becomes computational!

You will implement

- data audit;
- train/validation/test splits;
- classical baselines;
- neural networks;
- model comparison;
- failure analysis.

You will decide

- which representation is appropriate;
- which metric matters;
- whether the result is reliable;
- what astrophysical claim is justified.

References and further reading I

-  Cabrera-Vives et al. (2024). *ATAT: Astronomical Transformer for time series And Tabular data*. *Astronomy & Astrophysics*, 689, A289.
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References and further reading II

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-  Zhao et al. (2023). *Can diffusion model conditionally generate astrophysical images?*. Monthly Notices of the Royal Astronomical Society.
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-  Mudur and Finkbeiner (2022). *Can denoising diffusion probabilistic models generate astrophysical fields?*. NeurIPS ML4PhysicalSciences Workshop.
-  scikit-learn documentation. *Probability calibration and metrics*.
-  Angelopoulos and Bates (2021). *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification*.

A reliable astrophysical ML study should make two claims:

- ❶ **What the model predicts.**
- ❷ **Why that prediction supports a limited astrophysical conclusion.**

Do not confuse the first with the second.